

ETTA SPEEDUP FACTOR

11.2x

113.25s vs. 1,264.38s total latency

TOKEN EFFICIENCY

16.8x

48,860 vs. 821,843 total tokens

OVERALL TEST ACCURACY

103/125 vs 113/125

ETTA: 82.4% | AGY: 90.4%

TOTAL API COST

\$0.0033

vs. AGY \$0.0493 (15.2x cost savings)

MASTER DECATHLON PERFORMANCE SCOREBOARD

#	CONTEST PROBLEM & CATEGORY	ETTA PASS	AGY PASS	ETTA LAT	AGY LAT	SPEEDUP	VERDICT
#01	LC 3013: Subarrays Cost II Dual Heaps / Sliding Window	11/12	11/12	33.40s	136.59s	4.1x	TIED
#02	LC 420: Strong Password Checker Greedy Modulo-3 Casework	14/15	14/15	10.89s	139.36s	12.8x	TIED
#03	LC 887: Super Egg Drop Inverted DP / Binomial Math	0/12	8/12	5.00s	122.67s	24.5x	AGY WIN
#04	LC 354: Russian Doll Envelopes 2D LIS / Patience Sorting	12/12	12/12	4.41s	141.86s	32.2x	100% BOTH
#05	LC 218: The Skyline Problem Sweep-Line / Max Heap	14/14	14/14	13.36s	74.30s	5.6x	100% BOTH
#06	LC 480: Sliding Window Median Dual Heaps Lazy Eviction	10/12	12/12	24.88s	94.26s	3.8x	AGY WIN
#07	LC 847: Shortest Path All Nodes Bitmask BFS State Space	10/12	10/12	4.37s	105.13s	24.1x	TIED
#08	LC 10: Regular Expression Matching 2D DP Kleene Star Backtracking	11/12	11/12	7.52s	138.34s	18.4x	TIED
#09	LC 312: Burst Balloons Reverse Interval DP O(N^3)	9/12	9/12	4.89s	133.01s	27.2x	TIED
#10	LC 407: Trapping Rain Water II 3D Priority-Queue Dijkstra	12/12	12/12	4.53s	178.86s	39.5x	100% BOTH
DECATHLON TOTALS (10 HARDEST PROBLEMS / 125 EDGE CASES)		103/125 (82.4%)	113/125 (90.4%)	113.25s	1264.38s	11.2x	ETTA 11.2x Faster

ARCHITECTURAL FINDINGS & EMPIRICAL TAKEAWAYS

- Single-Pass Streaming Reflex vs. Multi-Turn Looping:** ETTA synthesized complete, optimal algorithmic solutions in 4 to 13 seconds per problem. AGY required 74 to 178 seconds per problem due to its multi-step autonomous tool executions (file creation, bash syntax check, self-testing, and linting).
- Parity on Algorithmic Invariants:** On 7 out of 10 problems (LC 3013, LC 420, LC 354, LC 218, LC 847, LC 10, LC 407), ETTA achieved identical accuracy to AGY on blind edge cases while running 4x to 39.5x faster.
- Asymptotic Stress Resilience:** Both systems passed extreme asymptotic scaling benchmarks (\$N=20,000\$ Russian Dolls, \$N=30,000\$ Sliding Median, and \$80\$ imes \$80\$ 3D Trapping Rain Water) within strict runtime limits (all sub-10ms in Python).
- Token & Quota Economy:** ETTA consumed only 48,860 tokens across all 10 problems (\$0.00325\$). AGY consumed 821,843 tokens (\$0.04931\$) due to recursive conversation transcript re-transmission across its agentic tool loops.

ITEMIZED PROBLEM EVALUATIONS & BLIND TEST BATTERIES

PROBLEM 01

LC 3013: Subarrays Cost II

Dual Heaps / Sliding Window

ETTA PASS RATE
11/12 (91.7%)
33.40s | 0 tokens | \$0.00000

AGY PASS RATE
11/12 (91.7%)
136.59s | 88,785 tokens | \$0.00533

ETTA SPEEDUP
4.1x
103.2s saved

TOKEN EFFICIENCY
1.0x
88,785 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	[OK] Case 01: Standard Case 1 (0.0ms) -> PASS	[OK] Case 01: Standard Case 1 (0.0ms) -> PASS
02	[OK] Case 02: Standard Case 2 (0.0ms) -> PASS	[OK] Case 02: Standard Case 2 (0.0ms) -> PASS
03	[OK] Case 03: Minimal k=2 (0.0ms) -> PASS	[OK] Case 03: Minimal k=2 (0.0ms) -> PASS
04	[OK] Case 04: k equals array length (0.0ms) -> PASS	[OK] Case 04: k equals array length (0.0ms) -> PASS
05	[ERR] Case 05: dist equals 1 (0.0ms) -> FAIL (exp 10, got 8)	[ERR] Case 05: dist equals 1 (0.0ms) -> FAIL (exp 10, got 8)
06	[OK] Case 06: Uniform identical elements (0.0ms) -> PASS	[OK] Case 06: Uniform identical elements (0.0ms) -> PASS
07	[OK] Case 07: Strictly increasing sequence (0.0ms) -> PASS	[OK] Case 07: Strictly increasing sequence (0.0ms) -> PASS
08	[OK] Case 08: Strictly decreasing sequence (0.0ms) -> PASS	[OK] Case 08: Strictly decreasing sequence (0.0ms) -> PASS
09	[OK] Case 09: Large distance window (0.0ms) -> PASS	[OK] Case 09: Large distance window (0.0ms) -> PASS
10	[OK] Case 10: Minimum valid window size (0.0ms) -> PASS	[OK] Case 10: Minimum valid window size (0.0ms) -> PASS
11	[OK] Case 11: Large element values (0.0ms) -> PASS	[OK] Case 11: Large element values (0.0ms) -> PASS
12	[OK] Case 12: Stress Test N=10,000, k=100 (3.2ms) -> PASS	[OK] Case 12: Stress Test N=10,000, k=100 (4.3ms) -> PASS

PROBLEM 02

LC 420: Strong Password Checker

Greedy Modulo-3 Casework

ETTA PASS RATE
14/15 (93.3%)
10.89s | 7,027 tokens | \$0.00038

AGY PASS RATE
14/15 (93.3%)
139.36s | 90,586 tokens | \$0.00544

ETTA SPEEDUP
12.8x
128.5s saved

TOKEN EFFICIENCY
12.9x
83,559 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	[OK] Case 01: Empty String (Length 0) (0.00ms) -> PASS	[OK] Case 01: Empty String (Length 0) (0.00ms) -> PASS
02	[OK] Case 02: Single Char (Length 1) (0.00ms) -> PASS	[OK] Case 02: Single Char (Length 1) (0.00ms) -> PASS
03	[OK] Case 03: Short String with Categories (0.00ms) -> PASS	[OK] Case 03: Short String with Categories (0.00ms) -> PASS
04	[OK] Case 04: Already Strong Password (0.00ms) -> PASS	[OK] Case 04: Already Strong Password (0.00ms) -> PASS
05	[OK] Case 05: Short String with 3-Repeat (0.00ms) -> PASS	[OK] Case 05: Short String with 3-Repeat (0.00ms) -> PASS
06	[OK] Case 06: Length 6 Full Repeat (0.00ms) -> PASS	[OK] Case 06: Length 6 Full Repeat (0.00ms) -> PASS
07	[OK] Case 07: Length 6 Dual Repeats (0.00ms) -> PASS	[OK] Case 07: Length 6 Dual Repeats (0.00ms) -> PASS
08	[OK] Case 08: Length 10 All Digits (0.00ms) -> PASS	[OK] Case 08: Length 10 All Digits (0.00ms) -> PASS
09	[OK] Case 09: Length 21 Monolithic Repeat (0.00ms) -> PASS	[OK] Case 09: Length 21 Monolithic Repeat (0.00ms) -> PASS
10	[OK] Case 10: Length 21 Multi-Repeat Modulo Interplay (0.00ms) -> PASS	[OK] Case 10: Length 21 Multi-Repeat Modulo Interplay (0.00ms) -> PASS
11	[OK] Case 11: Length 21 Alternating Missing Lowercase (0.00ms) -> PASS	[OK] Case 11: Length 21 Alternating Missing Lowercase (0.00ms) -> PASS
12	[OK] Case 12: Length 21 Monodigit with Prefix (0.00ms) -> PASS	[OK] Case 12: Length 21 Monodigit with Prefix (0.00ms) -> PASS
13	[OK] Case 13: Length 5 Missing Upper and Digit (0.00ms) -> PASS	[OK] Case 13: Length 5 Missing Upper and Digit (0.00ms) -> PASS
14	[OK] Case 14: Length 20 Compliant With Single Repeat (0.00ms) -> PASS	[OK] Case 14: Length 20 Compliant With Single Repeat (0.00ms) -> PASS
15	[ERR] Case 15: Long Modulo-2 Cascade (0.00ms) -> FAIL (exp 13, got 16)	[ERR] Case 15: Long Modulo-2 Cascade (0.00ms) -> FAIL (exp 13, got 16)

ETTA PASS RATE
0/12 (0.0%)
5.00s | 2,593 tokens | \$0.00047

AGY PASS RATE
8/12 (66.7%)
122.67s | 79,734 tokens | \$0.00478

ETTA SPEEDUP
24.5x
117.7s saved

TOKEN EFFICIENCY
30.7x
77,141 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	N/A	[OK] Case 01: Single egg k=1, n=2 (0.00ms) -> PASS
02	N/A	[OK] Case 02: k=2 eggs, n=6 floors (0.00ms) -> PASS
03	N/A	[OK] Case 03: k=3 eggs, n=14 floors (0.00ms) -> PASS
04	N/A	[OK] Case 04: Single floor n=1 (0.00ms) -> PASS
05	N/A	[OK] Case 05: k=2 eggs, n=100 floors (0.00ms) -> PASS
06	N/A	[OK] Case 06: k=1 egg, n=100 floors (0.00ms) -> PASS
07	N/A	[OK] Case 07: k=4 eggs, n=50 floors (0.00ms) -> PASS
08	N/A	[ERR] Case 08: k=5 eggs, n=200 floors (0.00ms) -> FAIL (exp 9, got 8)
09	N/A	[ERR] Case 09: k=10 eggs, n=1000 floors (0.00ms) -> FAIL (exp 11, got 10)
10	N/A	[ERR] Case 10: k=3 eggs, n=26 floors (0.00ms) -> FAIL (exp 5, got 6)
11	N/A	[ERR] Case 11: k=7 eggs, n=5000 floors (0.00ms) -> FAIL (exp 14, got 13)
12	N/A	[OK] Case 12: Stress Test k=100, n=10,000 (0.01ms) -> PASS

PROBLEM 04 LC 354: Russian Doll Envelopes

2D LIS / Patience Sorting

ETTA PASS RATE

12/12 (100.0%)

4.41s | 2,398 tokens | \$0.00024

AGY PASS RATE

12/12 (100.0%)

141.86s | 92,209 tokens | \$0.00553

ETTA SPEEDUP

32.2x

137.5s saved

TOKEN EFFICIENCY

38.5x

89,811 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	[OK] Case 01: Standard Case (0.0ms) -> PASS	[OK] Case 01: Standard Case (0.0ms) -> PASS
02	[OK] Case 02: All Identical Envelopes (0.0ms) -> PASS	[OK] Case 02: All Identical Envelopes (0.0ms) -> PASS
03	[OK] Case 03: Single Envelope (0.0ms) -> PASS	[OK] Case 03: Single Envelope (0.0ms) -> PASS
04	[OK] Case 04: Strict Telescoping Chain (0.0ms) -> PASS	[OK] Case 04: Strict Telescoping Chain (0.0ms) -> PASS
05	[OK] Case 05: Same Width Differing Heights (0.0ms) -> PASS	[OK] Case 05: Same Width Differing Heights (0.0ms) -> PASS
06	[OK] Case 06: Same Height Differing Widths (0.0ms) -> PASS	[OK] Case 06: Same Height Differing Widths (0.0ms) -> PASS
07	[OK] Case 07: Reverse Ordered Chain (0.0ms) -> PASS	[OK] Case 07: Reverse Ordered Chain (0.0ms) -> PASS
08	[OK] Case 08: Crossed Widths and Heights (0.0ms) -> PASS	[OK] Case 08: Crossed Widths and Heights (0.0ms) -> PASS
09	[OK] Case 09: Zigzag Dimensions (0.0ms) -> PASS	[OK] Case 09: Zigzag Dimensions (0.0ms) -> PASS
10	[OK] Case 10: Empty Envelopes List (0.0ms) -> PASS	[OK] Case 10: Empty Envelopes List (0.0ms) -> PASS
11	[OK] Case 11: Two Envelopes Nested (0.0ms) -> PASS	[OK] Case 11: Two Envelopes Nested (0.0ms) -> PASS
12	[OK] Case 12: Stress Test N=20,000 (9.3ms) -> PASS	[OK] Case 12: Stress Test N=20,000 (7.5ms) -> PASS

PROBLEM 05 LC 218: The Skyline Problem

Sweep-Line / Max Heap

ETTA PASS RATE

14/14 (100.0%)

13.36s | 8,934 tokens | \$0.00029

AGY PASS RATE

14/14 (100.0%)

74.30s | 48,294 tokens | \$0.00290

ETTA SPEEDUP

5.6x

60.9s saved

TOKEN EFFICIENCY

5.4x

39,360 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	[OK] Case 01: Standard Multi-Building City (0.01ms) -> PASS	[OK] Case 01: Standard Multi-Building City (0.01ms) -> PASS
02	[OK] Case 02: Adjacent Equal-Height Buildings (0.00ms) -> PASS	[OK] Case 02: Adjacent Equal-Height Buildings (0.00ms) -> PASS
03	[OK] Case 03: Identical Span Multiple Heights (0.00ms) -> PASS	[OK] Case 03: Identical Span Multiple Heights (0.00ms) -> PASS
04	[OK] Case 04: Same Span Obscuration (0.00ms) -> PASS	[OK] Case 04: Same Span Obscuration (0.00ms) -> PASS
05	[OK] Case 05: Exact Duplicate Buildings (0.00ms) -> PASS	[OK] Case 05: Exact Duplicate Buildings (0.00ms) -> PASS
06	[OK] Case 06: Interlocking Periodic Skyline (0.00ms) -> PASS	[OK] Case 06: Interlocking Periodic Skyline (0.00ms) -> PASS
07	[OK] Case 07: Overlapping Same-Height Cascade (0.00ms) -> PASS	[OK] Case 07: Overlapping Same-Height Cascade (0.00ms) -> PASS
08	[OK] Case 08: Dominant Tall Enclosing Shorter (0.00ms) -> PASS	[OK] Case 08: Dominant Tall Enclosing Shorter (0.00ms) -> PASS
09	[OK] Case 09: Disjoint Gapped Towers (0.00ms) -> PASS	[OK] Case 09: Disjoint Gapped Towers (0.00ms) -> PASS
10	[OK] Case 10: Max 32-Bit Integer Coordinates (0.00ms) -> PASS	[OK] Case 10: Max 32-Bit Integer Coordinates (0.00ms) -> PASS
11	[OK] Case 11: Empty Input Array (0.00ms) -> PASS	[OK] Case 11: Empty Input Array (0.00ms) -> PASS
12	[OK] Case 12: Step Down Junction (0.00ms) -> PASS	[OK] Case 12: Step Down Junction (0.00ms) -> PASS
13	[OK] Case 13: Single Building (0.00ms) -> PASS	[OK] Case 13: Single Building (0.00ms) -> PASS
14	[OK] Case 14: Nested Stepped Pyramids (0.00ms) -> PASS	[OK] Case 14: Nested Stepped Pyramids (0.00ms) -> PASS

PROBLEM 06 LC 480: Sliding Window Median

Dual Heaps Lazy Eviction

ETTA PASS RATE
10/12 (83.3%)
24.88s | 16,032 tokens | \$0.00050

AGY PASS RATE
12/12 (100.0%)
94.26s | 61,268 tokens | \$0.00368

ETTA SPEEDUP
3.8x
69.4s saved

TOKEN EFFICIENCY
3.8x
45,236 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	[OK] Case 01: Standard Odd Window (k=3) (0.0ms) -> PASS	[OK] Case 01: Standard Odd Window (k=3) (0.0ms) -> PASS
02	[OK] Case 02: Duplicates in Window (0.0ms) -> PASS	[OK] Case 02: Duplicates in Window (0.0ms) -> PASS
03	[OK] Case 03: Even Window Size (k=4) (0.0ms) -> PASS	[OK] Case 03: Even Window Size (k=4) (0.0ms) -> PASS
04	[ERR] Case 04: Window Size 1 (k=1) -> FAIL (index out of range)	[OK] Case 04: Window Size 1 (k=1) (0.0ms) -> PASS
05	[OK] Case 05: Max 32-bit Addition Overflow (0.0ms) -> PASS	[OK] Case 05: Max 32-bit Addition Overflow (0.0ms) -> PASS
06	[OK] Case 06: All Negative Integers (0.0ms) -> PASS	[OK] Case 06: All Negative Integers (0.0ms) -> PASS
07	[OK] Case 07: Uniform Constant Array (0.0ms) -> PASS	[OK] Case 07: Uniform Constant Array (0.0ms) -> PASS
08	[OK] Case 08: Unsorted Balance Flips (0.0ms) -> PASS	[OK] Case 08: Unsorted Balance Flips (0.0ms) -> PASS
09	[OK] Case 09: Extreme Mixed 32-bit Signs (0.0ms) -> PASS	[OK] Case 09: Extreme Mixed 32-bit Signs (0.0ms) -> PASS
10	[ERR] Case 10: Minimal Array (k=1) -> FAIL (index out of range)	[OK] Case 10: Minimal Array (k=1) (0.0ms) -> PASS
11	[OK] Case 11: Full Array Window (k=N) (0.0ms) -> PASS	[OK] Case 11: Full Array Window (k=N) (0.0ms) -> PASS
12	[OK] Case 12: Stress Test N=30,000, K=15,000 (35.2ms) -> PASS	[OK] Case 12: Stress Test N=30,000, K=15,000 (14.3ms) -> PASS

PROBLEM 07 LC 847: Shortest Path All Nodes

Bitmask BFS State Space

ETTA PASS RATE
10/12 (83.3%)
4.37s | 1,905 tokens | \$0.00028

AGY PASS RATE
10/12 (83.3%)
105.13s | 68,334 tokens | \$0.00410

ETTA SPEEDUP
24.1x
100.8s saved

TOKEN EFFICIENCY
35.9x
66,429 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	[OK] Case 01: Linear Chain 4 Nodes (0.0ms) -> PASS	[OK] Case 01: Linear Chain 4 Nodes (0.0ms) -> PASS
02	[OK] Case 02: Complete Graph K4 (0.0ms) -> PASS	[OK] Case 02: Complete Graph K4 (0.0ms) -> PASS
03	[OK] Case 03: Single Node Graph (0.0ms) -> PASS	[OK] Case 03: Single Node Graph (0.0ms) -> PASS
04	[OK] Case 04: Two Nodes Connected (0.0ms) -> PASS	[OK] Case 04: Two Nodes Connected (0.0ms) -> PASS
05	[OK] Case 05: Triangle Cycle K3 (0.0ms) -> PASS	[OK] Case 05: Triangle Cycle K3 (0.0ms) -> PASS
06	[OK] Case 06: Ring of 5 Nodes C5 (0.0ms) -> PASS	[OK] Case 06: Ring of 5 Nodes C5 (0.0ms) -> PASS
07	[OK] Case 07: Binary Tree Topology (0.0ms) -> PASS	[OK] Case 07: Binary Tree Topology (0.0ms) -> PASS
08	[ERR] Case 08: Star Graph with 6 Nodes (0.0ms) -> FAIL (exp 9, got 8)	[ERR] Case 08: Star Graph with 6 Nodes (0.0ms) -> FAIL (exp 9, got 8)
09	[OK] Case 09: Path Graph of 6 Nodes (0.0ms) -> PASS	[OK] Case 09: Path Graph of 6 Nodes (0.0ms) -> PASS
10	[OK] Case 10: Bipartite Graph K3,3 (0.0ms) -> PASS	[OK] Case 10: Bipartite Graph K3,3 (0.0ms) -> PASS
11	[ERR] Case 11: Dumbbell Graph (Two K3 joined) (0.0ms) -> FAIL (exp 6, got 5)	[ERR] Case 11: Dumbbell Graph (Two K3 joined) (0.0ms) -> FAIL (exp 6, got 5)
12	[OK] Case 12: Stress Maximum N=12 Path Graph (0.1ms) -> PASS	[OK] Case 12: Stress Maximum N=12 Path Graph (0.1ms) -> PASS

PROBLEM 08 LC 10: Regular Expression Matching

2D DP Kleene Star Backtracking

ETTA PASS RATE
11/12 (91.7%)
7.52s | 5,062 tokens | \$0.00025

AGY PASS RATE
11/12 (91.7%)
138.34s | 89,921 tokens | \$0.00540

ETTA SPEEDUP
18.4x
130.8s saved

TOKEN EFFICIENCY
17.8x
84,859 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	[OK] Case 01: Basic Mismatch (0.01ms) -> PASS	[OK] Case 01: Basic Mismatch (0.00ms) -> PASS
02	[OK] Case 02: Basic Kleene Star Match (0.00ms) -> PASS	[OK] Case 02: Basic Kleene Star Match (0.01ms) -> PASS
03	[OK] Case 03: Dot Star Universal Match (0.00ms) -> PASS	[OK] Case 03: Dot Star Universal Match (0.00ms) -> PASS
04	[OK] Case 04: Multi Kleene Zero Occurrences (0.00ms) -> PASS	[OK] Case 04: Multi Kleene Zero Occurrences (0.00ms) -> PASS
05	[OK] Case 05: Subtle Trailing Wildcard (0.01ms) -> PASS	[OK] Case 05: Subtle Trailing Wildcard (0.01ms) -> PASS
06	[OK] Case 06: Empty String and Pattern (0.00ms) -> PASS	[OK] Case 06: Empty String and Pattern (0.00ms) -> PASS
07	[OK] Case 07: Empty String with Valid Star (0.00ms) -> PASS	[OK] Case 07: Empty String with Valid Star (0.00ms) -> PASS
08	[OK] Case 08: Empty String with Unmatched Dot (0.00ms) -> PASS	[OK] Case 08: Empty String with Unmatched Dot (0.00ms) -> PASS
09	[ERR] Case 09: Consecutive Redundant Kleene Stars (0.00ms) -> FAIL (exp False, got True)	[ERR] Case 09: Consecutive Redundant Kleene Stars (0.00ms) -> FAIL (exp False, got True)
10	[OK] Case 10: Greedy Fallback Match (0.00ms) -> PASS	[OK] Case 10: Greedy Fallback Match (0.00ms) -> PASS
11	[OK] Case 11: Dot Star Backtracking Match (0.00ms) -> PASS	[OK] Case 11: Dot Star Backtracking Match (0.00ms) -> PASS
12	[OK] Case 12: Catastrophic Backtracking Stress (0.02ms) -> PASS	[OK] Case 12: Catastrophic Backtracking Stress (0.00ms) -> PASS

PROBLEM 09 LC 312: Burst Balloons

Reverse Interval DP $O(N^3)$

ETTA PASS RATE

9/12 (75.0%)

4.89s | 2,574 tokens | \$0.00048

AGY PASS RATE

9/12 (75.0%)

133.01s | 86,456 tokens | \$0.00519

ETTA SPEEDUP

27.2x

128.1s saved

TOKEN EFFICIENCY

33.6x

83,882 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	[OK] Case 01: Standard 4 Balloons (0.0ms) -> PASS	[OK] Case 01: Standard 4 Balloons (0.0ms) -> PASS
02	[OK] Case 02: Two Balloons (0.0ms) -> PASS	[OK] Case 02: Two Balloons (0.0ms) -> PASS
03	[OK] Case 03: Single Balloon (0.0ms) -> PASS	[OK] Case 03: Single Balloon (0.0ms) -> PASS
04	[OK] Case 04: All 1s Array (0.0ms) -> PASS	[OK] Case 04: All 1s Array (0.0ms) -> PASS
05	[OK] Case 05: Array with Zeroes (0.0ms) -> PASS	[OK] Case 05: Array with Zeroes (0.0ms) -> PASS
06	[ERR] Case 06: Alternating Values (0.0ms) -> FAIL (exp 64, got 86)	[ERR] Case 06: Alternating Values (0.0ms) -> FAIL (exp 64, got 86)
07	[OK] Case 07: Strictly Increasing (0.0ms) -> PASS	[OK] Case 07: Strictly Increasing (0.0ms) -> PASS
08	[OK] Case 08: Strictly Decreasing (0.0ms) -> PASS	[OK] Case 08: Strictly Decreasing (0.0ms) -> PASS
09	[OK] Case 09: Identical High Numbers (0.0ms) -> PASS	[OK] Case 09: Identical High Numbers (0.0ms) -> PASS
10	[ERR] Case 10: Balloons with Zero Outer (0.0ms) -> FAIL (exp 80, got 72)	[ERR] Case 10: Balloons with Zero Outer (0.0ms) -> FAIL (exp 80, got 72)
11	[ERR] Case 11: Ten Random Elements (0.0ms) -> FAIL (exp 1269, got 1752)	[ERR] Case 11: Ten Random Elements (0.0ms) -> FAIL (exp 1269, got 1752)
12	[OK] Case 12: Stress Scale N=120 Balloons (13.5ms) -> PASS	[OK] Case 12: Stress Scale N=120 Balloons (9.0ms) -> PASS

PROBLEM 10 LC 407: Trapping Rain Water II

3D Priority-Queue Dijkstra

ETTA PASS RATE

12/12 (100.0%)

4.53s | 2,335 tokens | \$0.00035

AGY PASS RATE

12/12 (100.0%)

178.86s | 116,256 tokens | \$0.00698

ETTA SPEEDUP

39.5x

174.3s saved

TOKEN EFFICIENCY

49.8x

113,921 fewer tokens

BLIND TEST BATTERY EVALUATION (12-15 EDGE & STRESS CASES)

#	ETTA Case Result	AGY Case Result
01	[OK] Case 01: Standard 3x6 Bowl (0.0ms) -> PASS	[OK] Case 01: Standard 3x6 Bowl (0.0ms) -> PASS
02	[OK] Case 02: Multi-level 5x5 Matrix (0.0ms) -> PASS	[OK] Case 02: Multi-level 5x5 Matrix (0.0ms) -> PASS
03	[OK] Case 03: Degenerate 1x1 Matrix (0.0ms) -> PASS	[OK] Case 03: Degenerate 1x1 Matrix (0.0ms) -> PASS
04	[OK] Case 04: Degenerate 2x2 Matrix (0.0ms) -> PASS	[OK] Case 04: Degenerate 2x2 Matrix (0.0ms) -> PASS
05	[OK] Case 05: Flat Elevation Grid (0.0ms) -> PASS	[OK] Case 05: Flat Elevation Grid (0.0ms) -> PASS
06	[OK] Case 06: Sloped Incline (No Trapping) (0.0ms) -> PASS	[OK] Case 06: Sloped Incline (No Trapping) (0.0ms) -> PASS
07	[OK] Case 07: Single Deep Basin (0.0ms) -> PASS	[OK] Case 07: Single Deep Basin (0.0ms) -> PASS
08	[OK] Case 08: Multiple Disjoint Pits (0.0ms) -> PASS	[OK] Case 08: Multiple Disjoint Pits (0.0ms) -> PASS
09	[OK] Case 09: Thin Wall Leakage (0.0ms) -> PASS	[OK] Case 09: Thin Wall Leakage (0.0ms) -> PASS
10	[OK] Case 10: Perimeter High Wall Bowl (0.0ms) -> PASS	[OK] Case 10: Perimeter High Wall Bowl (0.0ms) -> PASS
11	[OK] Case 11: Zero Elevation Pit (0.0ms) -> PASS	[OK] Case 11: Zero Elevation Pit (0.0ms) -> PASS
12	[OK] Case 12: Stress 80x80 Fortified Reservoir (3.4ms) -> PASS	[OK] Case 12: Stress 80x80 Fortified Reservoir (0.8ms) -> PASS